Linear Algebra (Y1S1) August 28, 2026

Linear Algebra: Concept Checklist

Active-Recall & Self-Testing Revision Guide

1 Systems of Linear Equations, Matrices, and Linear Transformations

[]**Conventions & Exam Pitfalls.**

- ▫ **Echelon Uniqueness:** The row echelon form (REF) of a matrix is ***not*** unique, but every matrix has a ***strictly unique*** reduced row echelon form (RREF).
- ▫ **Parametric Solutions:** When expressing the solution of an underdetermined consistent system, always express the ***pivot (basic) variables*** explicitly in terms of the ***free (non-basic) variables***.
- ▫ **Homogeneous Consistency:** A homogeneous linear system $Ax = 0$ is ***always consistent*** because it invariably admits the trivial solution $x = 0$.

[]Definitions & Concepts to State 1.1

- ▫ **Linear System Classifications:** Define a linear equation, a homogeneous linear equation, and a linear system. What does it mean for a system to be ***consistent*** versus ***inconsistent***?

- ▫ **Solution Cardinality:** State the 3 mutually exclusive cases for the solution set of any linear system over $\mathbb{R}$ or $\mathbb{C}$ (no solution, unique solution, infinitely many solutions).

- ▫ **Matrix Representations:** Given a linear system with m equations and n unknowns:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \quad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{cases}$$

  Write down its coefficient matrix, its column vector of constants b, and its augmented matrix $[A \mid b]$.

- ▫ **Elementary Row Operations:** List the three types of elementary row operations that preserve the solution set of a linear system.

- ▫ **Row Echelon Forms:** State the defining rules of:
  - Row Echelon Form (REF)
  - Reduced Row Echelon Form (RREF)

- ▫ **Elimination Algorithms:** Distinguish between ***Gaussian elimination*** (forward elimination to REF + back-substitution) and ***Gauss-Jordan elimination*** (reduction to RREF).

- ▫ **Variables Classification:** In an augmented matrix reduced to RREF:

$$\begin{bmatrix} 1 & 2 & 0 & 3 & 0 & 7 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{bmatrix}$$

  Identify the pivot columns, pivot (basic) variables, and free (non-basic) variables. Express the general solution.

- ▫ **Matrix Multiplication Definitions:** Let $A \in \mathbb{R}^{m \times r}$ and $B \in \mathbb{R}^{r \times n}$.
  - Write the index formula for the (i, j)-entry: $(AB)_{ij} = \sum_{k=1}^{r} a_{ik}b_{kj}$.
  - Express the i-th row of AB using the standard basis row vector e_iA.
  - Express the j-th column of AB using the standard basis vector Be_j.
  - Express the matrix-vector product Ax as a linear combination of the column vectors of A.

[]**Theorems & Identities to State 1.2**

- ▫ **Affine Line of Solutions:** If $S = (s_1, \ldots, s_n)$ and $N = (n_1, \ldots, n_n)$ are two distinct solutions to a linear equation (or consistent linear system) $Ax = b$, then for any scalar $t \in \mathbb{R}$, $(1-t)S + tN$ is also a valid solution.

- ▫ **Zero Row / Column Singularity:** If an $n \times n$ matrix A has an entire row or column of zeros, then A is singular (non-invertible).

- ▫ **RREF Dichotomy for Square Matrices:** If R is the RREF of an $n \times n$ matrix A, then either $R = I_n$ or R contains at least one row of zeros.

- ▫ **Algebraic Properties of Transpose and Trace:**
  - $(AB)^T = B^T A^T$
  - $\operatorname{tr}(A + B) = \operatorname{tr}(A) + \operatorname{tr}(B)$
  - $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ for conformable matrices A and B.

- ▫ **Invertibility Criteria & One-Sided Inverses:**
  - If A is invertible, its inverse A^{-1} is unique, and $(AB)^{-1} = B^{-1}A^{-1}$.
  - For square matrices of the same size, $AB = I \implies B = A^{-1}$ and $BA = I \implies B = A^{-1}$.

- ▫ **Symmetric Decomposition Theorem:** Any square matrix $A \in \mathbb{R}^{n\times n}$ can be uniquely decomposed as the sum of a symmetric matrix S and a skew-symmetric matrix K:

$$A = \underbrace{\frac{A + A^T}{2}}_{\text{Symmetric}} + \underbrace{\frac{A - A^T}{2}}_{\text{Skew-Symmetric}} .$$

- ▫ **Standard Matrix of Planar Rotation:** The standard matrix for the counter-clockwise rotation by an angle θ about the origin in $\mathbb{R}^2$ is:

$$[T] = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}.$$

- ▫ **Composition of Linear Transformations:** If $T_A : \mathbb{R}^m \to \mathbb{R}^k$ and $T_B : \mathbb{R}^k \to \mathbb{R}^n$, then the composition is a linear transformation with standard matrix $T_B \circ T_A = T_{BA}$.

[]**Proof Challenges & Calculations 1.3**

- ▫ **Infinite Solutions Proof:** Prove that if a linear system $Ax = b$ over $\mathbb{R}$ has at least two distinct solutions, it must have infinitely many solutions.
- ▫ **Trace Commutativity Proof:** Prove that for any $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times m}$, $\mathrm{tr}(AB) = \mathrm{tr}(BA)$.
- ▫ **Symmetric Products Proof:** For any rectangular matrix $A \in \mathbb{R}^{m\times n}$, prove that AA^T and A^TA are both symmetric matrices.
- ▫ **Rotation Matrix Derivation:** Derive the standard matrix for rotating a vector $(x, y) \in \mathbb{R}^2$ by an angle θ about the origin using trigonometry and basis vectors e_1, e_2.
- ▫ **Unit Basis Row/Column Extraction:** Let $e_i = [0, \ldots, 0, 1, 0, \ldots, 0]$ be the i-th standard basis vector.
  1. Prove that $e_iA = A_i$ (the i-th row of A).
  2. Prove that $Ae_j^T = A^j$ (the j-th column of A).

[]**Textbook Exercise Checklist.**

Section	**Textbook Problems**	**Status**
Sections 1.1–1.3	Problems 4, 5, 9, 19, 38–41, 45, 46, 53, 68, 74, 77–80, 84–86, 103	▫ Done
Sections 1.4–1.7	Problems 111, 123, 126, 127, 135, 162, 163, 172–175, 179–181, 212–216, 218, 220	▫ Done
Section 1.8	Problems 1, 2, 4, 12, 14, 22, 24, 26, 30, 32, 36, 46, 47	▫ Done

2 Determinants, Cofactor Expansion, and Cramer's Rule

[]**Definitions & Concepts to State 2.1**

- **Minors and Cofactors:** For an $n \times n$ matrix $A = [a_{ij}]$:
  - Define the minor M_{ij} of the entry a_{ij}.
  - Define the cofactor $C_{ij} = (-1)^{i+j} M_{ij}$.
- **Determinant Definition via Cofactor Expansion:** State the cofactor expansion formula along the i-th row and along the j-th column.
- **Triangular Matrices:** State the determinant formula for an upper triangular, lower triangular, or diagonal matrix.
- **Adjoint Matrix:** Define the adjoint (adjugate) matrix $\text{adj}(A)$ as the transpose of the cofactor matrix.
- **Cramer's Rule Formula:** Write down the explicit formula for solving $Ax = b$ using determinants when $\det(A) \neq 0$.

[]**Theorems & Identities to State 2.2**

- ▫ **Transpose Invariance:** $\det(A) = \det(A^T)$.
- ▫ **Elementary Matrix Determinants:** For any elementary matrix E, $\det(EA) = \det(E)\det(A)$.
- ▫ **Multiplicative Property of Determinants:** For any $n \times n$ matrices A and B:

$$\det(AB) = \det(A)\det(B).$$

- ▫ **Inverse Adjoint Formula:** If $\det(A) \neq 0$, then A is invertible with:

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A).$$

- ▫ **Invertible Matrix Theorem (IMT / Fundamental Equivalent Theorem):** State the core equivalences for an $n \times n$ matrix A:
  - A is invertible.
  - $\det(A) \neq 0$.
  - The RREF of A is I_n.
  - $Ax = 0$ has only the trivial solution $x = 0$.
  - $Ax = b$ is consistent with a unique solution for every $b \in \mathbb{R}^n$.
  - The column vectors (and row vectors) of A are linearly independent and span $\mathbb{R}^n$.

[]**Proof Challenges & Calculations 2.3**

- ▫ **Determinant of Elementary Products:** Prove that $\det(EA) = \det(E)\det(A)$ for each of the three types of elementary matrices E.
- ▫ **Multiplicative Determinant Proof:** Using elementary matrix factorizations of non-singular matrices, prove that $\det(AB) = \det(A)\det(B)$.
- ▫ **Cramer's Rule Derivation:** Derive Cramer's rule algebraically starting from the adjoint formula $x = A^{-1}b = \frac{1}{\det(A)} \operatorname{adj}(A)b$.

[]**Textbook Exercise Checklist.**

Section	Textbook Problems	Status
Sections 2.1–2.3	Problems 36, 61, 63, 65, 87, 93, 98, 104	□ Done

3 Euclidean Vector Spaces ($\mathbb{R}^n$)

[]**Conventions & Exam Pitfalls.**

- □ **Inner Product Matrix Form:** In $\mathbb{R}^n$, the dot product of two column vectors u, v is strictly equivalent to the matrix product $u^T v$.
- □ **Scalar vs. Vector Projection:** The scalar projection $\operatorname{comp}_a u$ is a scalar, whereas the vector projection $\operatorname{proj}_a u$ is a vector in the direction of a.

[]**Definitions & Concepts to State 3.1**

- □ **Vector Metrics in $\mathbb{R}^n$:** For vectors $u = (u_1, \ldots, u_n)$ and $v = (v_1, \ldots, v_n)$:
  - Define Euclidean norm (magnitude) $\|u\| = \sqrt{\sum_{i=1}^n u_i^2}$.
  - Define unit vector and the normalization process $u/\|u\|$.
  - Define Euclidean distance $d(u, v) = \|u - v\|$.
  - Define dot product $u \cdot v = \sum_{i=1}^n u_i v_i = u^T v$.
- □ **Geometric Angle and Orthogonality:** State the formula for the angle θ between non-zero vectors u, v, and define orthogonality ($u \perp v \iff u \cdot v = 0$).
- □ **Lines and Planes in $\mathbb{R}^n$:**
  - Point-normal equation of a hyperplane in $\mathbb{R}^n$: $n \cdot (x - x_0) = 0$.
  - Vector equation $r(t) = r_0 + tv$ and parametric equations of a line in $\mathbb{R}^n$.
  - Parameterization of a line segment connecting x_0 to x_1: $x(t) = (1-t)x_0 + tx_1$ for $t \in [0, 1]$.
- □ **Orthogonal Projections:** Define the orthogonal projection $\operatorname{proj}_a u$ of u along $a \neq 0$ and the orthogonal component vector $\operatorname{perp}_a u$.

[]**Theorems & Identities to State 3.2**

- **Cauchy-Schwarz Inequality:** For all $u, v \in \mathbb{R}^n$:

$$|u \cdot v| \leqslant \|u\|\|v\|,$$

  with equality holding if and only if u and v are collinear (linearly dependent).

- **Triangle Inequality:** For all $u, v \in \mathbb{R}^n$, $\|u+v\| \leqslant \|u\| + \|v\|$.
- **Parallelogram Law:** $\|u+v\|^2 + \|u-v\|^2 = 2(\|u\|^2 + \|v\|^2)$.
- **Polarization Identity:** $u \cdot v = \frac{1}{4}\left(\|u+v\|^2 - \|u-v\|^2\right)$.
- **Symmetric Matrix Inner Product Identity:** If $A \in \mathbb{R}^{n\times n}$ is symmetric ($A = A^T$), then for all $u, v \in \mathbb{R}^n$:

$$(Au) \cdot v = u \cdot (Av) = (Av) \cdot u.$$

- **Orthogonal Projection Theorem:** Let $a \neq 0$. Any vector u can be uniquely written as $u = w_1 + w_2$ where $w_1 = \operatorname{proj}_a u = \frac{u \cdot a}{\|a\|^2} a$ is parallel to a, and $w_2 = u - w_1$ is orthogonal to a.

[]**Proof Challenges & Calculations 3.3**

- **Geometric Identity Proofs:**
  1. Prove the Triangle Inequality using the Cauchy-Schwarz inequality.
  2. Prove the Parallelogram Law expanding the inner product norm definition.
  3. Prove that $(Au) \cdot v = u \cdot (Av)$ when A is symmetric.
- **Lines and Planes Calculations:**
  - Find the vector and parametric equations of the line in $\mathbb{R}^3$ passing through $(1, 2, -3)$ parallel to $(4, -5, 1)$.
  - Find the vector and parametric equations of the plane in $\mathbb{R}^3$ defined by $x - y + 2z = 5$.
- **Matrix Polynomial Inverse Challenge:** Let A be a square matrix satisfying $A^2 + A - 5I = 0$. Find an explicit expression for $(A + 2I)^{-1}$ in terms of A and I.

[]**Textbook Exercise Checklist.**

Section	**Textbook Problems**	**Status**
Chapter 3	Problems 73, 78, 82, 83, 88, 91, 96, 97, 101, 105, 112, 114, 116, 118, 121, 122, 125, 126, 131, 133, 135 (Cross product skipped)	□ Done

4 General Vector Spaces, Subspaces, and Fundamental Spaces

[]**Definitions & Concepts to State 4.1**

- **Vector Space and Subspace Axioms:** State the 10 vector space axioms and the 3-part test for a subset $W \subseteq V$ to be a ***subspace*** (non-empty, closure under addition, closure under scalar multiplication).
- **Span of a Set:** Define linear combination and span$(S) = \{\sum c_i v_i\}$.
- **Linear Independence:** Define linear independence and linear dependence for a set $S = \{v_1, \dots, v_r\}$. What if $S = \{v\}$ is a singleton set?
- **Wronskian of Functions:** Define the Wronskian $W(f_1, \dots, f_n)(x)$ of n differentiable functions.
- **Basis and Dimension:** Define a basis of a vector space V (spans V and linearly independent) and dimension $\dim(V)$.
- **Coordinates and Transition Matrices:**
  - Define the coordinate vector $(v)_B$ relative to a basis $B = \{u_1, \dots, u_n\}$.
  - Define the transition matrix $P_{B\to B'} = [(u_1)_{B'} \mid \cdots \mid (u_n)_{B'}]$ and state the relation $(v)_{B'} = P_{B\to B'}(v)_B$.
- **Fundamental Matrix Subspaces:** For an $m \times n$ matrix A:
  - Define Row space $\text{Row}(A) \subseteq \mathbb{R}^n$.
  - Define Column space $\text{Col}(A) \subseteq \mathbb{R}^m$.
  - Define Null space $\text{Null}(A) \subseteq \mathbb{R}^n$.
  - Define Rank ($\text{rank}(A) = \dim(\text{Col}(A)) = \dim(\text{Row}(A))$) and Nullity ($\text{nullity}(A) = \dim(\text{Null}(A))$).
- **Orthogonal Complements:** Define the orthogonal complement $W^\perp$ of a subspace $W \subseteq \mathbb{R}^n$.
- **Linear Transformation Kernel and Range:** For $T : V \to W$, define $\ker(T)$ and the range $R(T)$.

[]Theorems & Identities to State 4.2

- **Intersection of Subspaces:** If $W_1, \dots, W_r$ are subspaces of V, then $\bigcap_{i=1}^{r} W_i$ is also a subspace of V.
- **Null Space as Subspace:** The solution set of any homogeneous system $Ax = 0$ is a subspace of $\mathbb{R}^n$.
- **Minimal Subspace Property of Span:** $\mathrm{span}(S)$ is the smallest subspace of V containing S.
- **Linear Dependence Bound in $\mathbb{R}^n$:** If a set $S \subset \mathbb{R}^n$ has $r > n$ vectors, then S is linearly dependent.
- **Wronskian Independence Criterion:** If $W(f_1, \dots, f_n)(x_0) \neq 0$ for at least one $x_0 \in (a, b)$, then $\{f_1, \dots, f_n\}$ is linearly independent on (a, b).
- **Uniqueness of Basis Representation:** Let $B = \{v_1, \dots, v_n\}$ be a basis of V. Every vector $v \in V$ can be expressed as a linear combination of B in ***strictly one way***.
- **Equipotence of Bases:** All bases for a finite-dimensional vector space have the exact same number of vectors.
- **Transition Matrix Invertibility:** $P_{B \to B'}$ is always invertible with $P_{B \to B'}^{-1} = P_{B' \to B}$.
- **Consistency Theorem:** $Ax = b$ is consistent $\Leftrightarrow$ $b \in \mathrm{Col}(A)$.
- **Affine Structure of General Solutions:** If x_p is a particular solution to $Ax = b$, the general solution set is $x = x_p + x_h$ where $x_h \in \mathrm{Null}(A)$.
- **Invariance under Row Operations:** Elementary row operations preserve $\mathrm{Row}(A)$ and $\mathrm{Null}(A)$ (though they change $\mathrm{Col}(A)$).
- **Column Correspondence Principle:** Any linear dependency relation between the columns of A is identically preserved among the columns of any matrix B row-equivalent to A.
- **Dimension Theorem (Rank-Nullity):** For any $m \times n$ matrix A:

$$\mathrm{rank}(A) + \mathrm{nullity}(A) = n \quad \text{(number of columns)}.$$

- **Transpose Rank Invariance:** $\mathrm{rank}(A) = \mathrm{rank}(A^T) \leqslant \min(m, n)$.
- **Fundamental Subspaces Orthogonality:**
  - $\mathrm{Null}(A) = (\mathrm{Row}(A))^{\perp}$ in $\mathbb{R}^n$
  - $\mathrm{Null}(A^T) = (\mathrm{Col}(A))^{\perp}$ in $\mathbb{R}^m$

- **Kernel-Injectivity Equivalence:** A linear transformation T is one-to-one (in-

[]**Proof Challenges & Calculations 4.3**

- ▫ **Subspace Intersection Proof:** Prove that if W_1 and W_2 are subspaces of V, then $W_1 \cap W_2$ is a subspace of V.
- ▫ **Linear Dependence for $r > n$:** Prove that any set of r vectors in $\mathbb{R}^n$ with $r > n$ must be linearly dependent using homogeneous linear systems.
- ▫ **Wronskian Calculation:** Compute the Wronskian of $f_1(x) = e^x, f_2(x) = e^{2x}, f_3(x) = e^{3x}$ and prove they are linearly independent vectors in $C^\infty(\mathbb{R})$.
- ▫ **Basis Uniqueness Proof:** Prove the uniqueness of coordinates relative to a basis.
- ▫ **Transition Matrix Procedure:** Outline the augmented reduction method $[B' \mid B] \xrightarrow{\text{RREF}} [I \mid P_{B\to B'}]$ to compute transition matrices.
- ▫ **Zero-Product Rank Proof:** Prove that if $AB = 0$ for $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times s}$, then $\operatorname{rank}(A) + \operatorname{rank}(B) \leqslant n$ (Hint: show $\operatorname{Col}(B) \subseteq \operatorname{Null}(A)$).

[]**Textbook Exercise Checklist.**

Section	**Textbook Problems**	**Status**
Sections 4.1–4.3	Problems 34c, 37, 42, 45, 47, 48, 60, 64, 65	▫ Done
Sections 4.4–4.5	Problems 69, 74, 81, 82, 90, 95	▫ Done
Sections 4.6–4.8	Problems 108, 110, 113, 119, 125–127, 131–134, 137–139, 141, 144, 150, 153, 154, 158, 162	▫ Done
Sections 4.7–4.11	Problems 163, 222	▫ Done

5 Eigenvalues, Eigenvectors, and Diagonalization

[]Definitions & Concepts to State 5.1

- **Eigenvalue and Eigenvector Definitions:** For an $n \times n$ matrix A, define an eigenvalue λ and a corresponding non-zero eigenvector x such that $Ax = \lambda x$.
- **Characteristic Equation:** Define the characteristic polynomial $p(\lambda) = \det(\lambda I - A)$ and characteristic equation $\det(\lambda I - A) = 0$.
- **Eigenspace:** Define the eigenspace $E_\lambda = \mathrm{Null}(\lambda I - A)$ associated with an eigenvalue λ.
- **Matrix Similarity:** Define what it means for matrix B to be ***similar*** to matrix A ($B = P^{-1}AP$).
- **Diagonalizability:** Define what it means for a matrix A to be ***diagonalizable***.
- **Multiplicities:** Define the ***Algebraic Multiplicity*** (AM) and ***Geometric Multiplicity*** (GM) of an eigenvalue λ.

[]Theorems & Identities to State 5.2

- **Diagonalizability Characterization:** An $n \times n$ matrix A is diagonalizable $\Leftrightarrow$ A has n linearly independent eigenvectors.
- **Powers of Matrices:** If $Ax = \lambda x$, then for any positive integer k, $A^k x = \lambda^k x$. If $A = PDP^{-1}$, then $A^k = PD^kP^{-1}$.
- **Geometric Multiplicity Bound:** For every eigenvalue λ, $1 \leqslant \mathrm{GM}(\lambda) \leqslant \mathrm{AM}(\lambda)$.
- **Full Diagonalizability Criterion:** An $n \times n$ matrix A is diagonalizable over $\mathbb{R}$ $\Leftrightarrow$ all eigenvalues of A are real and $\mathrm{GM}(\lambda) = \mathrm{AM}(\lambda)$ for every eigenvalue λ.
- **Spectral Trace and Determinant Relations:**
  - $\mathrm{tr}(A) = \sum_{i=1}^{n} \lambda_i$
  - $\det(A) = \prod_{i=1}^{n} \lambda_i$

[]**Proof Challenges & Calculations 5.3**

- ▫ 2×2 **Eigensystem Calculation:** Find all eigenvalues and basis vectors for each eigenspace of:

$$A = \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix}.$$

- ▫ 3×3 **Matrix Diagonalization:** Find an invertible matrix P and a diagonal matrix D such that $P^{-1}AP = D$ for:

$$A = \begin{bmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix}.$$

- ▫ **Matrix Power Computation:** Outline the full procedure to compute A^k efficiently via diagonalization $A^k = PD^kP^{-1}$.
- ▫ **Eigenvalue Power Proof:** Prove by induction that if λ is an eigenvalue of A with eigenvector x, then λ^k is an eigenvalue of A^k with eigenvector x.

[]**Textbook Exercise Checklist.**

Section	**Textbook Problems**	**Status**
Chapter 5	Problems 3, 4, 5, 6, 7, 8, 13, 14, 16, 19, 28, 30–33, 34, 39, 45, 49, 51	▫ Done